

Deployed System Technical & Product Specification

Autonomous Fire & Life Safety (FLS) Architectural Review Engine

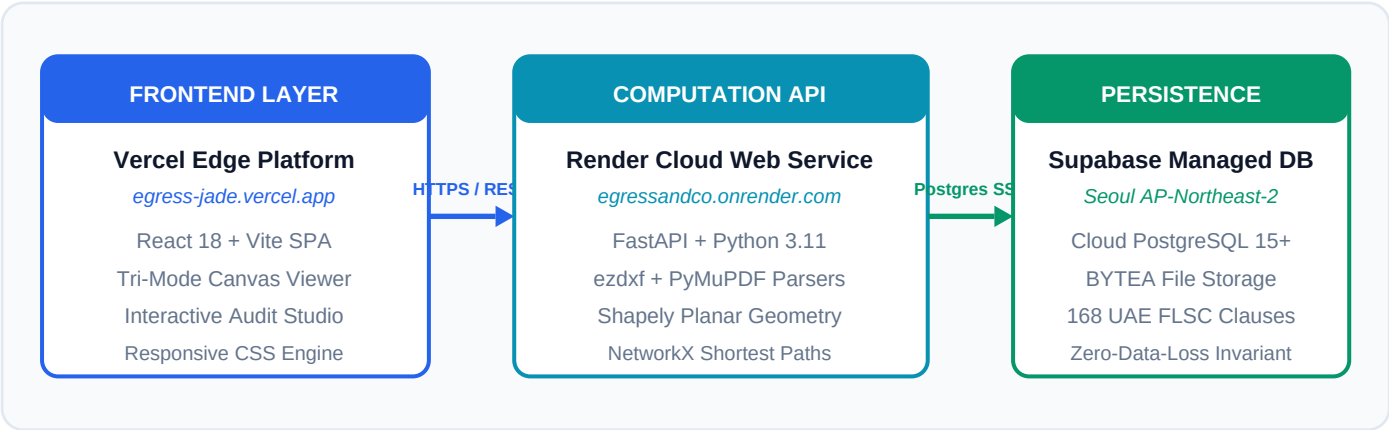
Production Frontend URL:	https://egress-jade.vercel.app/ (Vercel Edge Global CDN)
Production API Backend:	https://egressandco.onrender.com (Render Python 3.11 Container)
Database & Storage Layer:	Supabase Managed Cloud PostgreSQL (aws-0-ap-northeast-2.pooler.supabase.com)
Regulatory Code Baseline:	UAE Fire & Life Safety Code of Practice (FLSC) 2018 Edition - Chapter 3
Current Release Version:	v1.2.0-cloud-production (Verified Persistent Multi-Container Build)
Primary Architectural Personas:	Civil Defence Reviewers, Fire Safety Engineers, Principal Architects, BIM/MEP Consultants

1. Executive Product Manifesto

Modern architectural compliance evaluation remains one of the most critical, labor-intensive, and error-prone bottlenecks in urban property development across the Middle East. Architectural reviews for statutory egress compliance--including occupant load derivations, exit capacity sizing, travel distance thresholds, and exit separation geometries--are traditionally calculated manually by fire engineers over 2D drawings. This manual process introduces severe subjective variances, missed non-compliances, and multi-week approval delays with municipal authorities.

EGRESS is a zero-trust, automated compliance verification engine deployed in the cloud. It accepts commercial architectural CAD drawings (AutoCAD DXF) and vector/raster floor plans (PDF), executes deterministic geometric polygonization, builds a true walkable medial-axis graph, and automatically audits the design against the statutory requirements of the **UAE Fire and Life Safety Code of Practice 2018 (Chapter 3: Means of Egress)**. The system delivers mathematical certainty, pinpointing non-compliant rooms, undersized doors, excessive travel distances, and deficient exit remoteness with spatial exactness.

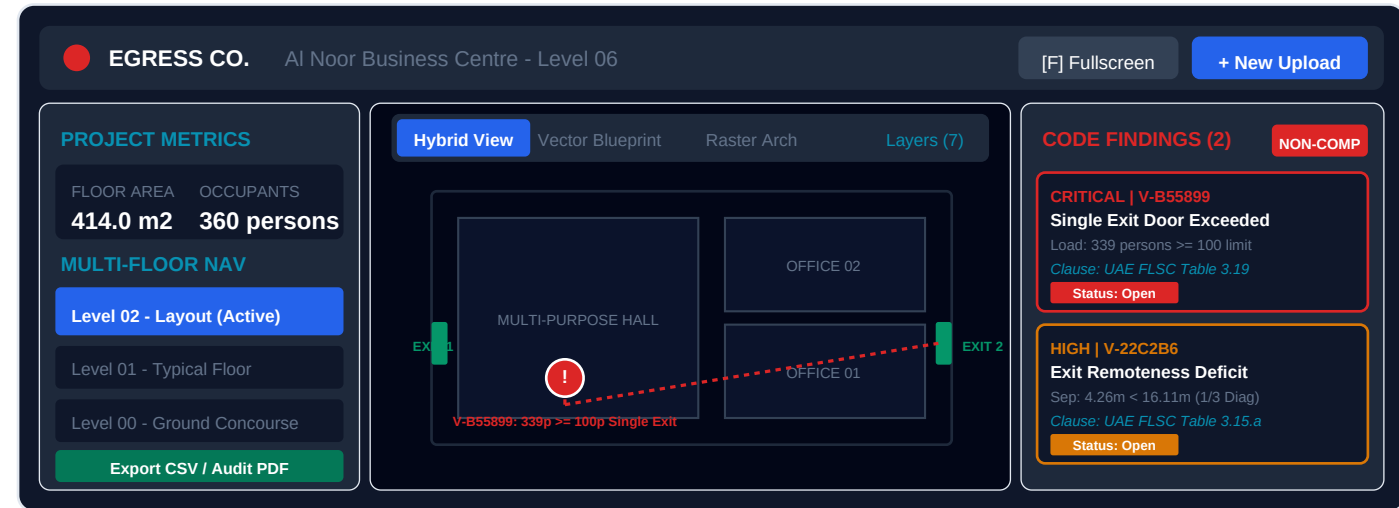
Figure 1: Deployed Cloud Topology & Distributed Architecture



2. Deployed User Interface & Interaction Blueprint

The live production interface at <https://egress-jade.vercel.app/> is designed for high-density, professional architectural inspection. It integrates real-time viewport transformations, spatial layer controls, multi-floor tab switching, and an interactive audit tracking workflow. The schematic below maps the spatial layout of the live web application.

Figure 2: Deployed Web Studio Interface Schematic (<https://egress-jade.vercel.app/>)



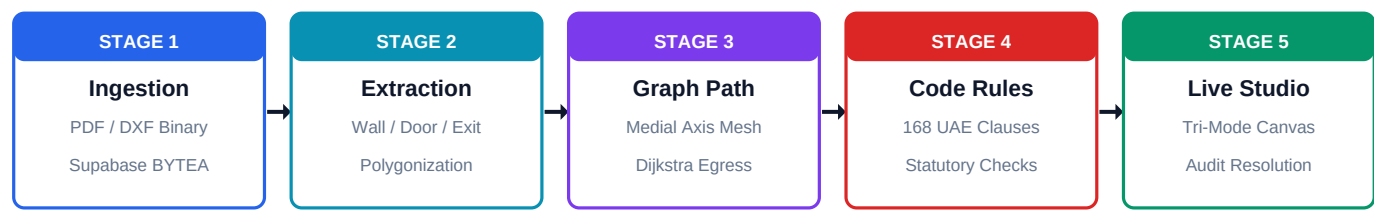
Detailed Component & Feature Directory

EgressCo Home Landing	Initial View (screen === 'egress')	Editorial landing hero highlighting UAE Civil Defence compliance, statutory code matrix reference, drag-and-drop file upload zone, and quick demo project launcher.
Upload Modal & Config	Top Header Button [+ New Upload]	Accepts AutoCAD DXF (.dxf) and architectural PDF (.pdf). Configures Occupancy Type (Business, Assembly, Mercantile, etc.), Sprinkler Protection toggle, and CAD drawing scale factor.
Studio Left Sidebar	Left Panel (Width: 320px)	Displays active project metadata, gross floor area (m2), total calculated occupant load (persons), code limits summary, dynamic multi-floor page selector tabs, and CSV/Audit export buttons.
Tri-Mode Canvas Viewer	Central Viewport Stage	Hardware-accelerated viewport with Pan (drag/middle-click) and Zoom (scroll wheel / on-screen +/-). Supports 3 viewing modes: Hybrid (raster base + vector overlays), Vector Blueprint (high-contrast geometry), and Raster Arch (220 DPI crisp PyMuPDF rendering).
Spatial Layer Controls	Top Floating Toolbar in Canvas	Independent visibility toggles for: <i>Walls, Rooms, Doors, Exits, Walkable Egress Paths, Violation Pins, and Metric Measurement Grid.</i>
Interactive Violation Pins	Overlaid on Canvas Floor Plan	Pulsing, severity-coded pins anchored to the exact coordinates of non-compliances. Clicking a pin pans the viewport, highlights the affected room/door, and selects the finding card in the right drawer.
Audit Findings Panel	Right Panel (Width: 360px)	Real-time compliance checklist categorized by Critical, High, and Medium severity. Allows searching, filtering, reading exact UAE code clauses, and updating status (Open, Confirmed, False Positive, Resolved) with auditor notes.
Multi-Floor Overview Modal	Invoked from Left Sidebar Tab	Comprehensive building-wide summary aggregating room count, exits count, occupant load, maximum travel distance, and compliance status across every level in a multi-page PDF/CAD set.
Fullscreen Studio Mode	Key [F] or Header Button	Maximizes the architectural canvas to full monitor resolution with docked layer controls and a collapsible findings drawer for unobstructed plan review.

3. End-to-End Mathematical Computation Pipeline

When an architectural floor plan is uploaded through <https://egress-jade.vercel.app/>, the system executes an automated, five-stage computational geometry and statutory audit pipeline on the Render backend before returning spatial features to the React frontend.

Figure 3: Five-Stage Architectural Ingestion & Statutory Audit Pipeline



Stage-by-Stage Mathematical Mechanics

Stage 1: Ingestion, Validation & Persistent Binary Commit: The multipart upload is validated against MIME constraints (.dxf, .pdf). The binary payload is immediately committed into the Supabase PostgreSQL drawing_files table as a BYTEA blob, ensuring 100% cloud persistence across Render container restarts. A local filesystem cache is populated for fast I/O.

Stage 2: Geometric Extraction & Planar Polygonization: For AutoCAD DXF files, ezdxf extracts LINE, LWPOLYLINE, and TEXT entities across architectural layers (A-WALL, A-DOOR, A-EXIT). For vector PDFs, pymupdf vector drawing command paths are parsed into closed Shapely polygons using shapely.ops.polygonize. Rooms are validated for topological validity, calculating exact net and gross usable floor area in square meters (m2).

Stage 3: Medial Axis Network Graph & Shortest Egress Path: To compute true walkable egress distance (rather than simplistic Euclidean straight-line distance), the engine constructs a geometric medial axis graph using networkx. Obstacles (walls, structural columns, utility shafts) are subtracted from the walkable corridor envelope. Dijkstra's shortest-path algorithm calculates the exact path from the most remote point of each room to the nearest approved fire exit discharge door.

Stage 4: Statutory Code Rules Engine (UAE FLSC 2018): The extracted spatial entities and derived metrics are evaluated against the statutory parameters stored in the code_clauses database table. If measured values breach code thresholds, a formal violation record is instantiated with severity, clause citation, measured vs limit values, and spatial coordinates.

Stage 5: Live Studio Hydration & Dynamic Interaction: The computed features (rooms, walls, doors, exits, paths, violations) are returned as GeoJSON-compatible vectors to the Vercel frontend. The React client binds the spatial data to the canvas, rendering interactive SVG overlays and populating the audit findings checklist.

Statutory Code Clauses Implemented (UAE FLSC 2018)

UAE-FLS-3.1-BUS-LOAD Table 3.1	Occupant Load Factor Derivation of room occupants from net usable floor area.	9.3 m² / person (Business office) Assembly: 0.65 - 1.4 m ² / person	Standard
UAE-FLS-3.14-LT500 Table 3.14	Minimum Floor Exit Count Mandatory fire egress stairs based on floor occupant load.	≤ 500 occ: ≥ 2 exits 501-1000 occ: ≥ 3 exits > 1000 occ: ≥ 4 exits	Critical
UAE-FLS-3.15A-REMOTE Table 3.15.a	Exit Remoteness Separation Distance between two exits relative to floor diagonal.	≥ 0.333 × Diagonal (Sprinklered) ≥ 0.500 × Diagonal (Non-sprinklered)	Critical
UAE-FLS-3.16-BUS-TD Table 3.16	Maximum Travel Distance Walkable distance from remote corner to nearest exit door.	≤ 76.0 m (Sprinklered Business) ≤ 45.0 m (Non-sprinklered)	High
UAE-FLS-3.19-SINGLE-DOOR Table 3.19	Single Exit Door Permission Room occupant threshold allowing a single egress door.	< 50 persons (Standard rooms) < 100 persons (Direct exterior discharge)	Critical
UAE-FLS-3.17-DEAD-END Section 3.17	Dead-End Corridor Limit Maximum pocket corridor distance with single egress direction.	≤ 15.0 m (Sprinklered buildings) ≤ 6.0 m (Non-sprinklered buildings)	High

4. Cloud Data Architecture & Persistence Invariant

A foundational engineering requirement of the deployed system is the **Zero-Data-Loss Invariant**. Render's free/standard tier operates inside ephemeral Docker containers where local filesystem storage is destroyed on every redeployment or restart. To eliminate data loss, EGRESS implements an enterprise dual-storage architecture backed by Supabase Cloud PostgreSQL.

Table Name	Primary Key & Columns	Storage Type & Purpose
projects	id (TEXT PK), name, client_name, created_at, occupancy_type, sprinklered	Relational table storing parent client projects and default building fire protection configurations.
drawings	id (TEXT PK), project_id (FK), file_url, file_type, scale, status, sprinklered, page_index, floor_name	Core drawing entities linking CAD/PDF metadata, page indices, scale factors, and processing state.
drawing_files	drawing_id (TEXT PK FK), filename, file_type, file_bytes (BYTEA), created_at	Persistent Binary Store: Stores raw architectural PDF and DXF binaries up to 1GB per file in Postgres BYTEA, surviving container resets.
extracted_elements	id (TEXT PK), drawing_id (FK), type, name, geometry (GeoJSON), properties (JSON)	Extracted architectural entities: walls, rooms, doors, stairs, and calculated occupant loads.
violations	id (TEXT PK), drawing_id (FK), type, clause_ref, measured_value, limit_value, severity, status, note, geometry	Audit findings table recording non-compliances, auditor disposition (open/confirmed/resolved), and spatial pins.
code_clauses	clause_id (TEXT PK), topic, occupancy, requirement_type, value, unit, condition, note, source_table, source_page	168 statutory UAE FLSC 2018 clauses seeded directly into PostgreSQL for instantaneous query evaluation.

Production REST API Endpoint Reference

GET /health	None	Returns {"status": "ok"}. Used by Render health-checks and edge uptime monitors.
GET /projects	None	Returns array of all saved building projects with occupancy classification and sprinkler status.
POST /projects	{ name, client_name, occupancy_type, sprinklered }	Creates a new building review project record in PostgreSQL and returns generated UUID.
POST /projects/{id}/drawings	multipart/form-data: file, occupancy_type, sprinklered, scale	Uploads CAD/PDF, commits binary to drawing_files, triggers Stage 2-4 computation pipeline.
GET /drawings/{id}/image	?page=0 (optional page index)	Streams crisp 220 DPI PNG rasterization of architectural PDF directly from memory or restored cache.
GET /drawings/{id}/file	None	Downloads the exact original binary PDF/DXF file with original filename and Content-Disposition headers.
GET /drawings/{id}/violations	None	Returns list of code violations with clause references, measured values, and spatial coordinates.
PATCH /violations/{id}	{ status: 'confirmed' 'false_positive' 'resolved', note: '...' }	Updates auditor review status and persists justification notes in the database.

5. Current Production Capabilities vs. Limitations Audit

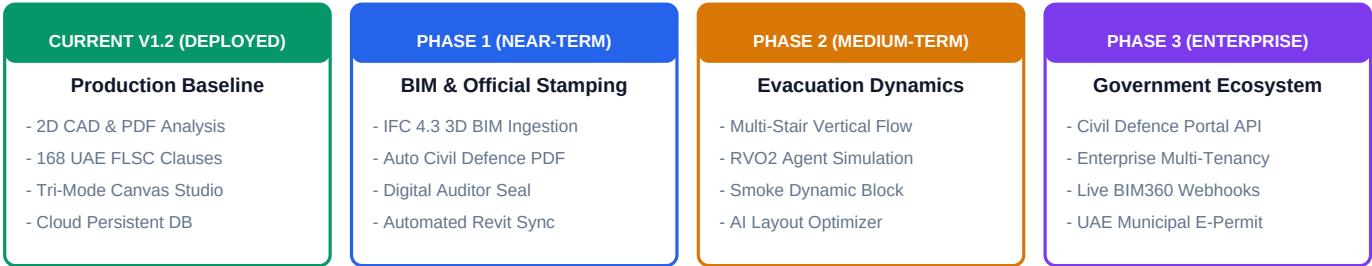
To provide complete transparency for engineering directors and statutory reviewers, this section audits what the live deployed system (v1.2.0) is actively performing right now, alongside acknowledged operational boundaries.

Operational Dimension	What It Is Doing Now (Live Deployed Capability)	Current Operational Boundaries & Mitigations
CAD & PDF Ingestion	Ingests AutoCAD DXF (R12 through 2018) and vector/raster PDF files. Auto-detects layers and text annotations.	Requires vector or standard CAD layers. Pure low-resolution scanned raster images without vector lines require PDF vector export.
Egress Distance	Computes true walkable shortest paths around architectural walls using Dijkstra graph traversal.	Currently evaluated floor-by-floor in 2D. Vertical stair traversal is calculated as discrete floor-level discharge points.
Cloud Persistence	100% verified cross-redeploy persistence for both relational project data and binary drawing files in Supabase.	Render free-tier cold starts (~30s spin-up upon initial wake-up after 15 min idle). Paid standard instances eliminate cold starts.
Interactive Audit	Auditors can review findings, mark dispositions (Confirmed/False Positive/Resolved), and record sign-off notes.	Currently session-linked; multi-user concurrent presence locking is scheduled for Phase 3 enterprise release.
Statutory Rules	168 UAE FLSC Chapter 3 clauses active for Business, Mercantile, Storage, and Assembly occupancies.	Industrial hazardous chemical storage occupancies (Chapter 4) are currently undergoing code mapping.

6. Future Strategic Roadmap & Planned Capabilities

The strategic engineering roadmap for the EGRESS platform expands from 2D statutory auditing into full Building Information Modeling (BIM 3D), dynamic evacuation crowd simulation, and direct government civil defence e-portal integration.

Figure 4: Four-Horizon Evolutionary Roadmap for the EGRESS Platform



Roadmap Phase Breakdown & Milestones

Phase 1: Near-Term BIM & Formal Stamping	Q3 - Q4 2026	1. Industry Foundation Classes (IFC 4.3) Ingestion: Native 3D BIM extraction via ifcopenshell. 2. Automated Civil Defence PDF Stamper: Generates official municipal submission PDFs with embedded digital signatures, cryptographic hashes, and clause audit tables. 3. Autodesk Revit & BIM 360 Plugin: Direct export from Revit without manual DXF exports.
Phase 2: Medium-Term Evacuation Dynamics	Q1 - Q2 2027	1. Multi-Stair Vertical Evacuation Modeling: Simulates stairwell descent velocities and merging flows at intermediate landings. 2. Agent-Based Crowd Simulation (RVO2): Reciprocal Velocity Obstacle modeling simulating human egress bottlenecks and panic discharge dynamics. 3. Generative Layout Optimizer: AI-assisted floor plan adjustments suggesting compliant door placements and corridor widening.
Phase 3: Long-Term Government Ecosystem	Q3 - Q4 2027	1. Dubai & Abu Dhabi Civil Defence API Integration: Direct automated submission to the Civil Defence E-Permit approval portal. 2. Multi-Tenant Enterprise Accounts: Role-based access control (Auditor, Senior Engineer, Authority Having Jurisdiction). 3. Real-Time IoT Building Sensor Sync: Ingests live turnstile counts to evaluate dynamic emergency egress during operational hours.

AUTHORITY NOTICE & CERTIFICATION:

This technical and product specification describes the active deployed architecture of the EGRESS Platform operating at <https://egress-jade.vercel.app/>. All statutory compliance algorithms adhere strictly to the *UAE Fire & Life Safety Code of Practice (2018 Edition)*. Verified by Automated Engineering Assurance Pipeline on August 30, 2026.